

Decoupled Image-to-Video Generation via Latent Flow: An Investigation into Embedding-Conditioned Rendering

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Abstract

001 *Image-to-video generation from a single static image and*
002 *an action label requires simultaneously understanding both*
003 *what is in the scene and how it should evolve over time.*
004 *We investigate a decoupled framework inspired by the Joint-*
005 *Embedding Predictive Architecture (JEPA): a causal Trans-*
006 *former (Flow Model) predicts the trajectory of CLIP em-*
007 *beddings in a compact 512-dimensional semantic space,*
008 *and a conditioned UNet (Renderer) converts each predicted*
009 *embedding back to a pixel frame. We conduct a system-*
010 *atic set of experiments across four Renderer architectures—*
011 *FiLM conditioning, cross-attention to CLIP patch tokens,*
012 *contrastive conditioning loss, and optical flow prediction—*
013 *and find that each variant collapses to a mode where the em-*
014 *bedding condition is effectively ignored and the output is a*
015 *near-identity transformation of the input frame. We present*
016 *a detailed failure analysis identifying the root causes: van-*
017 *ishing embedding motion signal between adjacent frames,*
018 *a training objective fundamentally misaligned with condi-*
019 *tioning sensitivity, and the inherent difficulty of encoding*
020 *spatial motion into a single global feature vector. Despite*
021 *the negative result, our analysis highlights key design chal-*
022 *lenges for embedding-conditioned video synthesis and sug-*
023 *gests concrete alternative approaches.*

024 1. Introduction

025 Image-to-video (I2V) generation aims to produce a tempo-
026 rally coherent video sequence from a single static frame.
027 When conditioned on an action label (e.g., “walking”), the
028 generated video should depict natural motion consistent
029 with that action. This task is relevant to content creation,
030 video editing, and synthetic data augmentation for action
031 recognition systems.

032 Recent large-scale diffusion models such as Stable Video
033 Diffusion and Wan2.2 have demonstrated impressive results
034 by inflating image diffusion architectures into the tempo-
035 ral domain. However, these end-to-end approaches incur

substantial computational costs and remain prone to motion
flickering and identity drift.

Inspired by the JEPA framework, we investigate an al-
ternative: decoupling the problem into a **Flow Model** that
predicts motion in a semantic embedding space and a **Ren-**
derer that converts semantic predictions back to pixels. The
key hypothesis is that motion dynamics are easier to model
in a compact, abstract embedding space than in raw pix-
els. In an ideal scenario, the Flow Model would learn high-
level action dynamics (“a walking person alternates legs”),
and the Renderer would faithfully translate these semantic
blueprints into frames.

We implement this framework using frozen CLIP ViT-
B/32 encoders, a causal Transformer for embedding trajec-
tory prediction, and a UNet-based Renderer with condition-
ing injected via FiLM, cross-attention, or explicit flow pre-
diction heads. We systematically ablate the conditioning
mechanism and loss function across four Renderer variants.

Our key finding is negative but instructive: across all
four variants, the Renderer learns to ignore the embedding
condition and produces near-identical outputs regardless of
input conditioning. The root cause is not any single archi-
tectural choice but a fundamental tension between (1) the
minimal variation in CLIP embeddings between adjacent
video frames, and (2) loss functions that penalize deviation
from a nearly-static ground truth.

We present a detailed failure analysis quantifying the
embedding signal strength, the collapse behavior across
architectures, and the training dynamics. We believe
these findings provide useful guidance for future work on
embedding-conditioned generative models.

2. Related Work

2.1. Image-to-Video Generation

Early video generation methods employed recurrent net-
works or generative adversarial networks. More recently,
diffusion-based approaches have become standard. Stable
Video Diffusion fine-tunes a pre-trained image diffusion
model on video data by adding temporal layers. Wan2.2 fol-

074 lows a similar approach at larger scale. These models gener-
075 ally operate in a VAE latent space, simultaneously modeling
076 spatial structure and temporal dynamics in one pipeline. We
077 experiment with a fundamentally different architecture that
078 attempts to factor these responsibilities across two modules.

079 2.2. Latent Predictive Architectures

080 The JEPA framework proposes that predictive learning
081 should operate in representation space rather than input
082 space. V-JEPA applies this principle to self-supervised
083 video learning by predicting masked spatio-temporal fea-
084 tures. Our Flow Model follows the same intuition: pre-
085 dict the trajectory of CLIP embeddings rather than raw pix-
086 els. Where our work differs is the downstream rendering
087 step—converting predicted embeddings back to pixels—
088 which proves to be the critical bottleneck.

089 2.3. Conditioned Image Generation

090 Feature-wise Linear Modulation (FiLM) and cross-
091 attention are standard techniques for conditioning image
092 generation on external information. FiLM applies channel-
093 wise affine transformations $\gamma(c) \odot x + \beta(c)$ predicted from
094 a conditioning vector c , and has been widely used in vi-
095 sual question answering, style transfer, and video predic-
096 tion. Cross-attention, popularized by the Transformer ar-
097 chitecture, allows each spatial position in a feature map to
098 attend to a set of conditioning tokens. Our experiments test
099 both mechanisms independently and in combination, and
100 find that neither is sufficient when the conditioning signal
101 (adjacent-frame CLIP embeddings) varies by less than 3%
102 in cosine distance.

103 2.4. Optical Flow and Warping

104 Frame interpolation and video prediction often employ ex-
105 plicit optical flow prediction followed by warping. Methods
106 such as RAFT achieve state-of-the-art optical flow estima-
107 tion through iterative refinement, while simpler approaches
108 use Farneback or TV-L1 flow as a supervision signal. Our
109 final Renderer variant adopts a flow-based approach: a
110 UNet backbone predicts a dense flow field conditioned on
111 CLIP embeddings, and the previous frame is warped via
112 differentiable bilinear sampling. We use Farneback optical
113 flow as pseudo-supervision and apply total variation regu-
114 larization to encourage spatial smoothness.

115 2.5. Action-Conditioned Video Generation

116 Motion generation conditioned on action labels has been
117 studied in controlled settings such as video game environ-
118 ments and simplified human motion benchmarks. Text-to-
119 video models such as Make-A-Video and Imagen Video
120 use large language model embeddings for open-vocabulary
121 conditioning at scale. Our setting is intermediate: we use

closed-set action labels encoded through CLIP text embed-
dings, providing a clean semantic signal in the same embed-
ding space as the visual features.

3. Data

We use the HMDB51 dataset, containing 6,754 video
clips across 51 human action categories such as “walking,”
“drinking,” “brushing hair,” and “climbing stairs.” Videos
average approximately 3 seconds in duration at 25–30 fps.
The dataset is split into training (70%) and test (30%) sets
following the first official split.

3.1. Preprocessing

For each video, we sample 17 consecutive frames ($n = 16$
prediction steps) at a stride of 2, and resize all frames to
 224×224 pixels. CLIP embeddings (CLS token + 49 patch
tokens) are pre-extracted for the entire dataset (242MB) to
avoid redundant forward passes during Flow Model train-
ing. The Renderer samples random frame pairs (I_{t-1}, I_t)
from each video with a gap of 1 frame.

3.2. Embedding Analysis

A key diagnostic we perform early is measuring the varia-
tion in CLIP embeddings across frames. Table 1 summa-
rizes pairwise cosine similarities.

Table 1. CLIP ViT-B/32 embedding cosine similarity. Same-video
frames are highly similar ($1 - \cos = 0.024$ for adjacent frames),
limiting the information available for frame-to-frame motion pre-
diction.

Comparison	Mean cos
Same video, distant frames ($\Delta t \approx 60$)	0.862
Same video, adjacent frames ($\Delta t = 1$)	0.976
Different videos, same action class	0.664
Different action classes	0.564

The critical observation is that adjacent-frame embed-
dings differ by only $1 - \cos \approx 0.024$ on average. This means
the embedding trajectory that the Flow Model must predict,
and that the Renderer must use as a conditioning signal,
contains very little per-frame variation. Any learning algo-
rithm must amplify this tiny signal by orders of magnitude
to produce meaningful pixel-space changes.

4. Methods

Our framework consists of three frozen encoders and two
trainable modules. Figure 1 illustrates the overall pipeline.

4.1. Encoders

Image Encoder. Frozen CLIP ViT-B/32. We extract both
the 512-dim CLS embedding E_t and the 49 spatial patch to-



Figure 1. Overall pipeline. Frozen CLIP encoders produce image embeddings E_t and label embeddings L . The Flow Model (causal Transformer) predicts $(\hat{E}_1, \dots, \hat{E}_n)$ from (L, E_0) . The Renderer (UNet with various conditioning mechanisms) produces each frame I_t from I_{t-1} conditioned on $\hat{E}_t$ and L .

kens (7×7 grid, 512-dim each) from the final vision transformer layer.

Label Encoder. Frozen CLIP text encoder. Action labels are converted from directory names to natural text (e.g., “brush hair”) and encoded to 512-dim semantic embeddings L .

4.2. Flow Model

A 4-layer causal Transformer encoder (12.9M parameters) with 8 attention heads and feedforward dimension 2048. The input sequence is $[L, E_0, E_1, \dots, E_{n-1}]$, and the model is trained with shifted teacher forcing: position t predicts $\hat{E}_t$ from $(L, E_{0,\dots,t-1})$.

We compare two loss functions:

- **MSE:** $\mathcal{L} = \frac{1}{n} \sum_t \|\hat{E}_t - E_t\|_2^2$
- **Cosine:** $\mathcal{L} = \frac{1}{n} \sum_t (1 - \cos(\hat{E}_t, E_t))$

4.3. Renderer

The Renderer is a UNet (4 down/up blocks, 40–43M parameters depending on variant) that takes I_{t-1} as input and produces an output conditioned on $\hat{E}_t$, L , and patch tokens. We experiment with four variants:

1. **FiLM (baseline):** channel-wise Feature-wise Linear Modulation from a condition vector $[\hat{E}_t; L] \in \mathbb{R}^{1024}$, applied at each residual block. Output: pixel residual ΔI_t . Loss: $L1(\Delta I_t, I_t - I_{t-1})$.
2. **FiLM + Cross-Attention:** FiLM conditioning plus multi-head cross-attention from UNet feature maps to CLIP patch tokens at the bottleneck and first two upsampling blocks. Output: pixel residual ΔI_t . Loss: L1.
3. **FiLM + Cross-Attention + Contrastive:** same as variant 2, plus a contrastive conditioning loss: for a batch, we compute $\mathcal{L}_{\text{cont}} = \|\cos(\Delta I_{\text{real}}, \Delta I_{\text{fake}}) - \cos(E_{\text{real}}, E_{\text{fake}})\|_2^2$, where fake conditions are shuffled embeddings from other videos in the batch. This explicitly penalizes the Renderer for producing identical outputs from different conditions.
4. **Flow Prediction:** same UNet backbone with FiLM + cross-attention, but the output head predicts a dense optical flow field $\in \mathbb{R}^{2 \times 224 \times 224}$. The previous frame is warped via the predicted flow: $I_t = \text{warp}(I_{t-1}, \text{flow})$.

$$\text{Loss: } L1(I_t^{\text{warped}}, I_t^{\text{gt}}) + \lambda_{\text{TV}} \cdot \text{TV}(\text{flow}) + \lambda_{\text{FB}} \cdot L1(\text{flow}, \text{Farneback}).$$

4.4. Training Details

All models are trained with AdamW ($lr = 10^{-4}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-2}) and a cosine annealing learning rate schedule over 100 epochs. We use a batch size of 8 with gradient accumulation over 2 steps (effective batch size 16). Gradient clipping with maximum norm 1.0 is applied. All experiments were conducted on a single NVIDIA RTX 4060 Laptop GPU (8GB VRAM). Complete training of the Flow Model takes approximately 4 hours; each Renderer variant takes approximately 7 hours.

Flow Model training data pipeline. Because passing CUDA models through PyTorch DataLoader workers is not supported on Windows, we pre-extract CLIP embeddings for all 6,754 videos (17 frames each) and save them to disk as a single 242MB file. The Flow Model training loop loads pre-extracted embeddings directly, requiring no CLIP forward passes during training. This eliminates the DataLoader CUDA issue and accelerates training by approximately $3 \times$.

Renderer training data pipeline. The Renderer uses HMDB51FrameDataset, which loads raw frames from disk (via OpenCV) and performs image transformations in the DataLoader workers. CLIP encoding happens in the main training loop where CUDA is available. Farneback optical flow is computed on CPU using OpenCV, adding approximately 30% overhead per batch.

4.5. Implementation Notes

Shifted teacher forcing. An earlier version of the Flow Model used “zero-placeholder” inputs where the model received $[L, E_0, \mathbf{0}, \dots, \mathbf{0}]$ and was expected to predict all future embeddings in a single forward pass. This created a train-inference mismatch: at inference time, the model sees its own predictions rather than zeros. Shifted teacher forcing, where the model receives the real embedding sequence and predicts one step ahead at each position, aligns training and inference exactly. We found this improved embedding trajectory quality significantly (from 2.0% to 82.5% of GT motion, combined with the cosine loss change).

Multi-scale flow prediction. In the flow prediction Renderer variant, we considered predicting flow at multiple decoder resolutions and combining them via coarse-to-fine refinement (similar to RAFT). However, initial experiments showed that even a single-scale flow head struggled to produce consistent non-zero outputs, so we deferred multi-scale investigation to future work.

5. Experiments

5.1. Evaluation Protocol

We evaluate using the following metrics:

- **Embedding Trajectory MSE:** MSE between predicted and ground-truth CLIP embedding sequences.
- **Inter-frame 1-cos Distance:** $1 - \cos(\hat{E}_t, \hat{E}_{t+1})$, measuring how much “motion” the predicted embeddings contain.
- **Conditioning Sensitivity:** $\frac{\|\text{Renderer}(I, E_t) - \text{Renderer}(I, E_0)\|_1}{\|\text{Renderer}(I, E_0)\|_1 + \epsilon}$, measuring whether changing the embedding changes the Renderer output.

5.2. Flow Model Results

Table 2 summarizes the Flow Model results.

Table 2. Flow Model performance with MSE vs. cosine similarity loss. Cosine loss prevents embedding collapse while maintaining prediction accuracy.

Metric	MSE Loss	Cosine Loss
Train loss (final)	0.0116	0.0264
Inter-frame 1-cos (GT)	0.0290	0.0290
Inter-frame 1-cos (pred)	0.0006	0.0239
% of GT motion preserved	2.0%	82.5%

The MSE-trained Flow Model achieves low MSE (0.0116) but collapses to predicting nearly identical embeddings at every timestep ($1 - \cos \approx 0.0006$). Cosine loss resolves this: predicted embeddings retain 82.5% of the ground-truth motion variation while maintaining comparable embedding quality (MSE between predicted and GT embeddings actually improves in terms of cosine distance).

5.3. Renderer Results

Table 3 presents the core negative result: across all four architectures, the Renderer learns to ignore the embedding condition.

For all variants, the conditioning sensitivity ratio is approximately 1.0, meaning the magnitude of the Renderer’s output does not meaningfully change when the conditioning embedding changes. In variants 1–3 (residual prediction), the absolute output magnitude converges to ≈ 0.005 —essentially near-zero. In variant 4 (flow prediction), the pre-

Table 3. Renderer conditioning sensitivity across four architecture variants. A ratio of ~ 1.0 indicates the model produces identical outputs regardless of the embedding condition.

Renderer Variant	Sensitivity Ratio
1. FiLM + L1	1.03x
2. FiLM + Cross-Attn + L1	0.98x
3. FiLM + Cross-Attn + L1 + Contrastive	0.98x
4. FiLM + Cross-Attn + Flow Pred + Rec + TV + FB	1.01x

dicted flow field converges back to zero after initial non-zero activity from random initialization.

5.4. Diagnostic Experiments

To isolate the failure mode, we conduct several diagnostic tests:

Embedding distance test. We feed the Renderer embeddings from frames at varying temporal distances ($[t, t + 1, t + 4, t + 8, t + 16]$) and measure the output magnitude. Even with frame $t + 16$ ($1 - \cos = 0.31$, a meaningful change), the output magnitude remains unchanged from the $t + 1$ case ($1 - \cos = 0.03$). This confirms that the model is not merely limited by the weak signal in adjacent frames—it has learned to ignore the embedding entirely.

Farneback flow as pseudo-target. We compute Farneback optical flow between adjacent frames (mean magnitude 0.39 px, max 2.84 px) and use it as a supervised target. The flow head initially produces non-zero flow but converges back to zero as the reconstruction and smoothness losses dominate.

CLIP feature discrimination. We verify that CLIP embeddings *do* contain frame-to-frame information: the cosine similarity gap between same-video adjacent frames (0.976) and different videos of the same class (0.664) is 0.312—a substantial difference. The information exists, but translating it from a 512-dim global vector into spatially meaningful pixel changes proves fundamentally difficult with our architectures.

6. Discussion: Why Did It Fail?

We identify four interrelated factors contributing to the Renderer’s failure:

6.1. Factor 1: Vanishing Embedding Gradient

Adjacent-frame CLIP embeddings differ by $1 - \cos \approx 0.024$. For the Flow Model, this is a learnable signal (we achieve 82.5% of GT variation). But for the Renderer, this means the conditioning vector $[E_t; L]$ changes by $< 3\%$ between frames—far too little for the UNet to learn a meaningful mapping from condition to output.

310	6.2. Factor 2: Loss Function Misalignment	358
311	All our loss functions (L1 on residuals, L1 on reconstruction after warping) have the property that the <i>optimal constant prediction</i> is already near the ground truth: $\Delta I_t \approx 0$	359
312	and flow ≈ 0 . The model can achieve a respectably low	360
313	loss by ignoring the condition and outputting a near-identity	361
314	transformation. The contrastive conditioning loss in variant	362
315	3 was designed to address this, but it only constrains the	363
316	<i>direction</i> of output variation, not its magnitude. A constant-	364
317	near-zero output with slightly different noise patterns per	365
318	condition can satisfy the contrastive constraint while still	366
319	producing no meaningful motion.	367
320		368
321		369
322	6.3. Factor 3: FiLM Cannot Express Spatial Motion	370
323	Feature-wise Linear Modulation applies channel-wise scal-	371
324	ing and shifting that is uniform across all spatial positions.	372
325	This can adjust global brightness or contrast but cannot	373
326	communicate “the hand should move to the upper-right	374
327	while the background stays still.” Cross-attention to CLIP	375
328	patch tokens (variant 2) was an attempt to introduce spatial	376
329	conditioning, but the patch tokens from CLIP ViT capture	377
330	<i>what</i> is in each region, not <i>how it should move</i> .	378
331		379
332		380
333	6.4. Factor 4: Representation Mismatch	381
334	CLIP embeddings are trained for image-text alignment;	382
335	they encode semantic content (“a person walking”) but not	383
336	spatial configuration (“the right leg is bent at 30 degrees and	384
337	positioned at coordinates (x, y)”). The information needed	385
338	to predict pixel-level frame changes requires spatial repre-	386
339	sentations that CLIP’s global pooling discards. Using patch	387
340	tokens helps but does not fully resolve this, as CLIP patch	388
341	tokens are not trained for dense correspondence tasks.	389
342		390
343		391
344	6.5. Limitations of Our Study	392
345	We acknowledge several limitations: (1) we only explored	393
346	CLIP ViT-B/32 as the embedding backbone—larger ViT	394
347	variants or dense-prediction models like DINOv2 may have	395
348	different properties; (2) we did not attempt joint end-to-end	396
349	training of the Flow Model and Renderer, which could al-	397
350	low the Flow Model to learn embeddings specifically opti-	398
351	mized for rendering; (3) we used a relatively small dataset	399
352	(6.7k videos), and the collapse behavior may differ at larger	400
353	scale where more diverse motion patterns provide a stronger	401
354	learning signal; (4) HMDB51 videos are constrained to spe-	402
355	cific human action categories, and the learned dynamics	403
356	may not generalize to more open-ended video generation.	404
357		405
358		406
359	6.6. What We Learned	407
360	Despite the negative result, this investigation yielded several	408
361	lessons that may benefit future work in this area:	409
362	Embedding choice is critical. The choice of CLIP ViT-	
363	B/32 as our embedding backbone was motivated by its joint	
364	vision-language space and compact dimensionality. How-	
365	ever, CLIP embeddings are optimized for semantic dis-	
366	crimination, not spatial correspondence. For a decoupled	
367	I2V pipeline to succeed, the embedding space must encode	
368	both <i>what</i> is in the scene and <i>where</i> things are spatially—	
369	properties that CLIP’s contrastive training objective does	
370	not guarantee.	
371	Loss functions define the optimization landscape. We	
372	observed that all tested pixel-space loss functions (L1, L1	
373	+ TV, L1 + Farneback) contain a near-identity trivial so-	
374	lution ($\Delta I \approx 0$, flow ≈ 0). This is not specific to our	
375	architecture—any model trained with a pixel reconstruction	
376	loss on adjacent-frame pairs faces the same challenge. Loss	
377	functions that operate in a learned feature space (perceptual	
378	loss, adversarial loss) may provide a stronger signal for mo-	
379	tion learning.	
380	The JEPA philosophy needs a rendering pathway.	
381	JEPA’s core insight—predict in representation space, not	
382	input space—proved effective for our Flow Model (82.5%	
383	of GT motion). But the companion challenge of <i>decod-</i>	
384	<i>ing</i> those predictions back to the pixel domain is equally	
385	hard and received comparatively little attention in our im-	
386	plementation. The rendering problem deserves its own care-	
387	ful treatment rather than being treated as an architectural	
388	afterthought.	
389	6.7. Future Work	
390	Several promising directions emerge from this analysis.	
391	Dense spatial embeddings. Replacing CLIP with	
392	a model trained for dense visual correspondence—such	
393	as DINOv2, Stable Diffusion’s VAE latents, or a self-	
394	supervised optical flow network—could provide embed-	
395	dings that naturally encode spatial configuration. A tra-	
396	jectory in DINOv2 patch feature space would carry richer	
397	spatial information than a trajectory in CLIP CLS space.	
398	Joint end-to-end training. Training the Flow Model	
399	and Renderer jointly, with gradients flowing from the ren-	
400	dering loss back through the Flow Model, would allow the	
401	embedding space to adapt to the needs of the Renderer.	
402	The Flow Model would no longer be constrained to match	
403	CLIP’s embedding structure and could learn features specifi-	
404	cally optimized for frame rendering.	
405	Iterative refinement. Rather than predicting the full	
406	frame transformation in a single forward pass, a multi-step	
407	refinement process (similar to diffusion models or recurrent	
408	refinement) could gradually build up the output from an ini-	
409	tial near-identity estimate. This would allow the model to	
410	learn motion in stages—coarse structure first, then fine de-	
411	tails.	
412	Adversarial and perceptual losses. GAN-based losses	
413	or CLIP-based perceptual losses directly penalize lack of	
414	motion and unrealistic static output, potentially breaking	
415	the near-identity local minimum that trapped all of our L1-	

410 trained models.

411 **Explicit motion field supervision.** Using a pre-trained

412 optical flow network (e.g., RAFT) instead of Farneback as

413 a supervision target could provide a cleaner learning signal.

414 Combined with a rendering loss, this could guide the model

415 toward physically plausible motion while maintaining vi-

416 sual fidelity.

417 **7. Conclusion**

418 We set out to build a decoupled image-to-video genera-

419 tion system in which a causal Transformer predicts action-

420 conditioned embedding trajectories and a conditioned UNet

421 renders those trajectories to pixels. The Flow Model com-

422 ponent was partially successful: with cosine similarity

423 loss it learned to predict embedding trajectories preserving

424 82.5% of ground-truth motion variation, while MSE loss

425 led to complete collapse. However, the Renderer failed

426 across all four conditioning architectures tested: FiLM,

427 cross-attention, contrastive conditioning, and optical flow

428 prediction.

429 Our analysis suggests the failure is not due to any sin-

430 gle architectural choice but stems from a fundamental mis-

431 match: the Renderer is asked to extract spatially-precise

432 motion instructions from a conditioning signal that varies

433 by less than 3% between frames and that was never de-

434 signed to encode spatial configuration. All tested loss func-

435 tions contain a near-identity trivial solution that dominates

436 the optimization.

437 **Future directions** suggested by this analysis include:

438 (1) replacing the global CLIP embedding with dense spatial

439 features from a model trained for correspondence (e.g., DI-

440 NOv2, optical flow networks); (2) training the Flow Model

441 and Renderer jointly end-to-end, so the Flow Model learns

442 to produce embeddings that are directly useful for render-

443 ing rather than matching a pre-trained representation; (3)

444 replacing the single-step rendering approach with an iter-

445 ative or diffusion-based process where the model can re-

446 fine its output over multiple steps; (4) using adversarial or

447 perceptual losses that penalize the absence of motion rather

448 than L1 deviation from a static ground truth.

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